

A Causal Multi-State Framework for Biomarker-Guided Dialysis Initiation

Optimizing Treatment Timing using Restricted Residual Mean Survival Time (RRMST)

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Background: Chronic Kidney Disease (CKD)

What is Chronic Kidney Disease?

- A progressive condition where the kidneys gradually lose their ability to filter waste from the blood.
- Disease progression is closely monitored using time-varying biomarkers, primarily the estimated Glomerular Filtration Rate (eGFR).

The End Stage:

- When eGFR drops dangerously low, patients reach kidney failure.
- At this stage, life-sustaining renal replacement therapy—**Dialysis**—is required to survive.

The Clinical Dilemma: The Research Problem

When is the optimal time to start dialysis?

This is one of the most challenging decisions in nephrology:

- **Start Too Early:** The patient is subjected to the severe physical toll, infections, and lifestyle burdens of dialysis before it is strictly necessary.
- **Wait Too Long:** The patient risks fatal complications (e.g., heart failure, severe toxicity) and may die while waiting.

The Problem

Currently, there is no personalized, dynamic rule. We want to predict the optimal timing to initiate dialysis based on a patient's **entire accumulated clinical history up to today**, rather than just a single cross-sectional snapshot.

Why is this difficult? (Statistical Challenges)

Standard statistical models (like basic survival analysis) fail here because of three major hurdles:

- ❶ **Competing Risks:** A patient might die from CKD complications *before* they ever get the chance to start dialysis.
- ❷ **Immortal Time Bias:** Patients who start dialysis later naturally had to survive long enough to reach that point, which can make delayed treatment look artificially better.
- ❸ **Confounding by Indication:** Doctors start dialysis *because* the patient is getting sicker. If we just look at raw data, dialysis looks like it causes death, because only the sickest patients receive it.

Our Goal and Framework

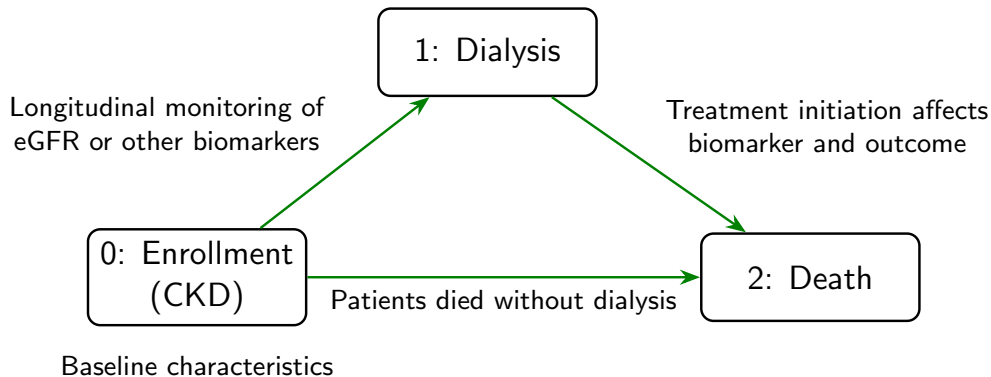
Our Goal:

To build a statistical framework that can tell a doctor, at any given clinic visit: *“Based on this patient’s entire longitudinal history up to today, will they live longer if we start dialysis immediately, or if we defer for δ months?”*

How we represent the patient journey:

We model the patient’s journey through a **Multi-State Model**. At any time, a patient is in one of three states.

The 3-State Disease Pathway



Notation: Tracking the Data

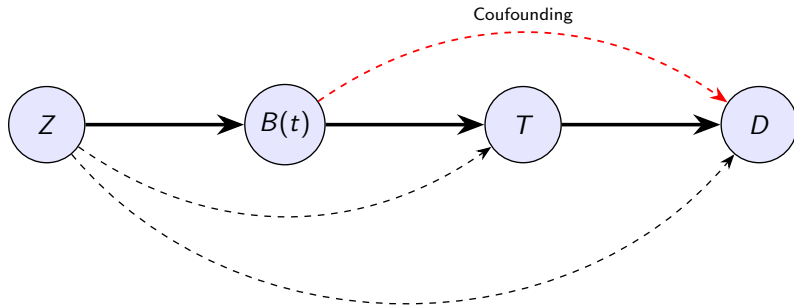
To make this mathematically rigorous, we track four key pieces of information for every patient i :

- Z_i : **Baseline Characteristics** (e.g., age, sex, genetics at enrollment).
- $B_i(u)$: **Longitudinal Biomarkers** (e.g., the continuous trajectory of their eGFR over time).
- $T_{i,A}$: **Treatment Time** (When they move from State 0 to State 1).
- D_i : **Death Time** (When they enter State 2).

At any clinical decision point (which we call a **Landmark**, t_k), we summarize the patient's **entire accumulated history** up to that day as $H_i(t_k)$.

The Causal Diagram (DAG)

How do all these variables interact? The diagram below reveals why finding the true effect of dialysis is so difficult:



- The biomarker $B(t)$ dictates the Treatment (T), but $B(t)$ also directly causes Death (D). This is a classic causal confounding loop (Confounding by Indication).

Bridging the Gap: Core Causal Assumptions

To trust that our predicted life expectancy (Δ) represents a *true causal effect*, our data must satisfy strict identifying conditions:

- **No Unmeasured Confounding (Sequential Exchangeability):** We must capture *all* variables driving the doctor's decision. If doctors initiate dialysis based on unrecorded acute symptoms, our model remains biased.
- **Positivity and Consistency:** We only evaluate this rule when there is a genuine choice (clinical equipoise). Also, $D_i = D_i^g$ if the observed treatment path follows regime g .
- **Independent Censoring:** $D_i^g \perp\!\!\!\perp C_i \mid H_i(u)$ for $u \geq t_k$.

Mathematical Identification: The Foundation

How do we formally link the unobservable causal world to observable data?

1. The Eligible Population: At landmark t_k , we evaluate patients alive and untreated:

$$\mathcal{E}_k = \{D_i > t_k, T_{i,A} > t_k\}$$

2. The Target Estimand ($S^g(r)$): The expected counterfactual survival:

$$S^g(r) = \mathbb{E}[Y^g(r) \mid \mathcal{E}_k] = \int \mathbb{E}[Y^g(r) \mid H_k = h, \mathcal{E}_k] dF_{H_k \mid \mathcal{E}_k}(h)$$

3. Time-Indexed Sequential Exchangeability:

$$Y^g(r) \perp\!\!\!\perp dA(u) \mid \mathcal{H}(u), \bar{A}(u^-) = \bar{g}(u^-)$$

At every moment u , the actual treatment decision $dA(u)$ is independent of future counterfactual survival, conditional on the evolving clinical history $\mathcal{H}(u)$.

Mathematical Identification: The Multi-State G-Formula

Under consistency and independent censoring, the causal survival is identified by manipulating the observed transition hazards.

For a fixed deferral regime g_δ (defer for δ months):

$$S_i^{g_\delta}(r \mid b_i^{(m)}) = \exp \left\{ - \int_0^{\min(r, \delta)} \lambda_{02}(t_k + a \mid b_i^{(m)}) da \right\} \\ \times \exp \left\{ - \int_0^{\max(0, r - \delta)} \lambda_{12}(s \mid v = t_k + \delta, b_i^{(m)}) ds \right\}$$

Structurally Eliminating Immortal Time Bias

By explicitly forcing the simulated patient to remain untreated for δ months, they mathematically accumulate the pre-dialysis mortality risk (λ_{02}). A rapidly failing patient is assigned a high probability of death *before* they ever reach the delayed treatment time $t_k + \delta$.

What is G-Computation?

Introduced by James Robins (1986), **g-computation** is a mathematical engine that simulates a randomized trial using biased, observational data.

It works in three steps:

- ① **Fit (Observe)**: Model how the disease naturally progresses and how events naturally happen using multi-state hazard models.
- ② **Intervene (Force)**: Take a simulated patient and mathematically *force* them to follow a specific rule (g). This severs the confounding loop.
- ③ **Integrate (Average)**: Calculate the expected survival under that forced rule by product-integrating the observational hazards.

The Final Output: Causal RRMST

Standard models give us **Hazard Ratios** (e.g., $HR = 0.8$), which are hard to interpret and lack a causal definition in multi-state systems.

By integrating the g-computation survival curves over a specific time horizon ($\Delta\tau$), we translate complex math into a universally understood clinical metric: **Restricted Residual Mean Survival Time (RRMST)**.

$$\text{RRMST}_i^g(t_k, \Delta\tau \mid H_i(t_k)) = \int_0^{\Delta\tau} \hat{S}_i^g(r) dr$$

The Decision Rule

Our framework outputs the estimated expected months of survival gained over the restricted horizon:

$$\Delta = \text{RRMST}(\text{Start Now}) - \text{RRMST}(\text{Defer})$$

The Joint Multi-State Causal Model

We build upon standard joint models (e.g., Ferrer, 2016) by defining distinct continuous and intervention-driven hazard structures using **Shared Random Effects** (b_i).

1. Longitudinal Sub-Model (eGFR Trajectory):

$$m_i(u) = \beta_0 + \beta_1 u + \gamma^\top Z_i + b_{i,0} + b_{i,1} u$$

2. Multi-State Hazard Sub-Models:

- **Pre-Dialysis ($0 \rightarrow 1$ & $0 \rightarrow 2$):** Depend *continuously* on the concurrent biomarker level $m_i(u)$ and slope $m'_i(u)$ at any specific time u .
- **Post-Dialysis ($1 \rightarrow 2$):** Dialysis alters the disease mechanism. This hazard depends on baseline covariates, a function of the dialysis initiation time (v), and the biomarker *frozen* exactly at initiation $m_i(v)$.

1. Model Specification: Pre-Dialysis Hazards

Estimation Strategy: We fit a **Full-Cohort Joint Model** from absolute time $u = 0$ (enrollment). Causal g-computation later predicts conditionally on landmark eligibility.

Pre-Dialysis Transitions ($0 \rightarrow 1$ and $0 \rightarrow 2$):

Hazards depend continuously on the concurrent latent trajectory $m_i(u) = X_i(u)^\top \beta + Z_i(u)^\top b_i$.

Note: The longitudinal model restricts Y_i to pre-dialysis eGFR.

$$\lambda_{01}(u \mid b_i) = \lambda_{01,0}(u) \exp\{\alpha_{01}^\top Z_i + \phi_{01,1} m_i(u) + \phi_{01,2} m'_i(u)\}$$

$$\lambda_{02}(u \mid b_i) = \lambda_{02,0}(u) \exp\{\alpha_{02}^\top Z_i + \phi_{02,1} m_i(u) + \phi_{02,2} m'_i(u)\}$$

2. Model Specification: Post-Dialysis Hazard

Post-Dialysis Transition ($1 \rightarrow 2$):

Uses a clock-reset scale ($s = u - v$). The hazard depends on the *frozen* trajectory at treatment time v and a linear delay penalty:

$$\lambda_{12}(s \mid v, b_i) = \lambda_{12,0}(s) \exp\{\alpha_{12}^\top Z_i + \theta_1 m_i(v) + \theta_2 m'_i(v) + \psi_1 v\}$$

Note on Delay ($\psi_1 v$): The post-dialysis delay effect is estimated as a function of absolute dialysis initiation time v . For causal regimes at a fixed landmark t_k , we substitute $v = t_k$ (g_0) or $v = t_k + \delta$ (g_δ).

Baseline Hazards ($\lambda_{hk,0}$): Modeled as **piecewise-constant** over a fixed calendar grid (e.g., every 6 months). For numerical stability and guaranteed positivity during BFGS optimization, we parameterize on the log-scale: $\lambda_{hk,0}(t) = \exp(\xi_{hk,j})$ for $t \in [c_{j-1}, c_j]$.

Time Origins and Likelihood Contributions

The survival likelihood $L_{S,i}$ depends on the patient's observed full-cohort event pattern:

- **Case A (Dialysis at v , Death at D_i):**

$$\lambda_{01}(v) \exp \left\{ - \int_0^v (\lambda_{01} + \lambda_{02}) du \right\} \times \lambda_{12}(D_i - v \mid v) \exp \left\{ - \int_0^{D_i - v} \lambda_{12}(s \mid v) ds \right\}$$

- **Case B (Dialysis at v , Censored at C_i):**

$$\lambda_{01}(v) \exp \left\{ - \int_0^v (\lambda_{01} + \lambda_{02}) du \right\} \times \exp \left\{ - \int_0^{C_i - v} \lambda_{12}(s \mid v) ds \right\}$$

- **Case C (Death at D_i before Dialysis):**

$$\lambda_{02}(D_i) \exp \left\{ - \int_0^{D_i} (\lambda_{01} + \lambda_{02}) du \right\}$$

- **Case D (Censored at C_i Untreated):**

$$\exp \left\{ - \int_0^{C_i} (\lambda_{01} + \lambda_{02}) du \right\}$$

Solving the Model: Joint Maximum Likelihood

We account for measurement error and event-dependent biomarker observation under the assumed joint model.

1. Joint Likelihood Formulation:

- The likelihood combines the linear mixed model density and the multi-state transition probabilities, integrated over the random effects b_i .

2. Estimation Strategy:

- Maximize the joint likelihood using the Expectation-Maximization (EM) algorithm.
- Integrals over b_i are solved via pseudo-adaptive Gauss-Hermite quadrature.

The Advantage

This strictly propagates the uncertainty of the eGFR trajectories into the survival estimates and provides the model-based **joint posterior distribution** of b_i for each patient.

3. The Expected Complete Data Likelihood (E-Step)

The complete data is $\mathcal{C}_i = \{Y_i, T_{i,A}, D_i, C_i, \delta_{T,i}, \delta_{D,i}, b_i\}$. At iteration (it), we compute the expected log-likelihood $Q(\theta \mid \theta^{(it)})$:

$$Q(\theta \mid \theta^{(it)}) = \sum_{i=1}^N \int [\log f(Y_i \mid b_i; \theta_y) + \log L_{S,i}(\theta_t, \beta \mid b_i) + \log f(b_i; \theta_b)] f(b_i \mid \mathcal{C}_i; \theta^{(it)}) db_i$$

Numerical Integration Strategies:

- **Over Random Effects (db_i):** Adaptive Gauss-Hermite Quadrature. We will test $K = 7, 9, 11$ points to ensure stability as random-effect dimensionality increases.
- **Over Time (du, ds):** Robust **grid integration** (e.g., Gauss-Legendre integration). We will conduct sensitivity checks to guard against numerical-integration criticism.

4. The ECM-Step (Expectation Conditional Maximization)

Unlike basic joint models, our M-step **does not cleanly separate**. Because the latent trajectory $m_i(u) = X_i(u)^\top \beta + Z_i(u)^\top b_i$ drives the transition hazards, the longitudinal fixed effects (β) appear in *both* the longitudinal and survival likelihoods.

We utilize an **ECM** strategy:

- **Update θ_b (D) and σ^2 :** Conditionally updated via closed-form solutions (weighted empirical covariance/variance).
- **Update β and θ_t ($\lambda_0, \alpha, \phi, \psi, \theta$):** Updated jointly (or conditionally alternating) using numerical optimization (e.g., **BFGS** quasi-Newton algorithm).

5. Convergence and The Truncated Posterior

Convergence Criterion: $\max_j |\theta_j^{(it+1)} - \theta_j^{(it)}| < 10^{-5}$

Crucial Distinction for G-Computation (Stage 2):

For model fitting, the E-step uses the full-data posterior. For causal decision-making at landmark t_k , we **cannot peek into the future**. We draw exclusively from the **truncated landmark posterior**:

$$P(b_i | Y_i(t_k), Z_i, T_{i,A} > t_k, D_i > t_k) \propto f(Y_i(t_k) | b_i) \exp \left\{ - \int_0^{t_k} [\lambda_{01}(u | b_i) + \lambda_{02}(u | b_i)] du \right\} f(b_i)$$

Uncertainty Reporting: For simulation efficiency, we will initially report bias, RMSE, and empirical SD. Full parameter uncertainty ($\hat{\theta}$ via block bootstrap) combined with trajectory uncertainty (b_i via Monte Carlo) will be reserved for a smaller clinical demonstration.

Primary Estimator: G-Computation Survival Integrals

We mathematically force the regimes and integrate the hazards to obtain the counterfactual survival curves $S^g(r)$:

1. Immediate Initiation (g_0):

Patient transitions immediately ($v = t_k$):

$$S_i^{g_0}(r | b_i^{(m)}) = \exp \left\{ - \int_0^r \lambda_{12}(s | v = t_k, b_i^{(m)}) ds \right\}$$

2. Forced Deferral for δ Months (g_δ):

Patient runs the gauntlet of pre-dialysis risk for δ months, then transitions ($v = t_k + \delta$):

$$\begin{aligned} S_i^{g_\delta}(r | b_i^{(m)}) = & \exp \left\{ - \int_0^{\min(r, \delta)} \lambda_{02}(t_k + a | b_i^{(m)}) da \right\} \\ & \times \exp \left\{ - \int_0^{\max(0, r - \delta)} \lambda_{12}(s | v = t_k + \delta, b_i^{(m)}) ds \right\} \end{aligned}$$

The Challenge (Drift Bias):

Freezing covariates at t_k assumes eGFR stops declining during the deferral period, which mathematically underestimates pre-dialysis mortality.

The Solution: Monte Carlo Marginalization

- Draw M latent trajectories $b_i^{(m)}$ from the **truncated landmark posterior**: $P(b_i \mid Y_i(t_k), Z_i, T_{i,A} > t_k, D_i > t_k)$ to prevent peeking into the future.
- Evaluate the estimated mathematical hazards continuously along each updated path.
- Average the resulting survival curves to integrate out biological uncertainty:

$$\hat{S}_i^g(r) = \frac{1}{M} \sum_{m=1}^M S_i^g(r \mid b_i^{(m)})$$

Algorithm Summary: The Decision Engine

Algorithm 1 Joint Multi-State G-Computation

Require: Pre-dialysis history $Y_i(t_k)$, covariates Z_i , horizon $\Delta\tau$, deferral δ , draws M .

- 1: **Stage 1: Joint Model Estimation (Full Cohort)**
- 2: Fit joint model linking LMM and multi-state hazards via ECM algorithm.
- 3: Extract truncated landmark posterior $P(b_i | Y_i(t_k), Z_i, T_{i,A} > t_k, D_i > t_k)$.
- 4: **Stage 2: Path-Integrated Microsimulation**
- 5: **for** $m \leftarrow 1$ **to** M **do**
- 6: Sample latent trajectory $b_i^{(m)}$ from the truncated posterior.
- 7: **Regime** g_0 (**Start Now**, $v = t_k$):
- 8: Compute $S_i^{g_0}(r | b_i^{(m)}) = \exp \left\{ - \int_0^r \lambda_{12}(s | v = t_k, b_i^{(m)}) ds \right\}$
- 9: **Regime** g_δ (**Defer δ Months**, $v = t_k + \delta$):
- 10: Compute $S_i^{g_\delta}(r | b_i^{(m)}) = \exp \left\{ - \int_0^{\min(r, \delta)} \lambda_{02}(t_k + a | b_i^{(m)}) da \right\} \times \exp \left\{ - \int_0^{\max(0, r - \delta)} \lambda_{12}(s | v = t_k + \delta, b_i^{(m)}) ds \right\}$
- 11: **end for**
- 12: Marginalize: $\hat{S}_i^g(r) \leftarrow \frac{1}{M} \sum_{m=1}^M S_i^g(r | b_i^{(m)})$ for $g \in \{g_0, g_\delta\}$.
- 13: **Stage 3: Causal Decision**
- 14: $\text{RRMST}^g \leftarrow \int_0^{\Delta\tau} \hat{S}_i^g(r) dr$ for $g \in \{g_0, g_\delta\}$.
- 15: $\Delta \leftarrow \text{RRMST}^{g_0} - \text{RRMST}^{g_\delta}$
- 16: **return Initiate Dialysis** if $\Delta > 0$ (In the simulation, we use $\Delta > 0$. In clinical deployment, this can be replaced by $\Delta > c$, where c reflects a clinically meaningful survival gain or quality-of-life tradeoff.), else **Defer**.

Longitudinal Biomarker (eGFR Proxy)

Observed every 3 months with measurement error:

$$Y_{ij} = m_i(t_{ij}) + \varepsilon_{ij}, \quad \varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2)$$

$$m_i(t) = \underbrace{\beta_0 + \beta_1 t + \gamma Z_i}_{\text{fixed effects}} + \underbrace{b_{i0} + b_{i1} t}_{\text{random effects}}, \quad \mathbf{b}_i \sim N(\mathbf{0}, \mathbf{D})$$

Parameter	Description	True value
β_0	Population intercept	40.0
β_1	Population slope (per month)	-1.00
γ	Confounder effect on intercept	-2.00
σ_{b0}^2	Random intercept SD	0.50
σ_{b1}^2	Random slope SD	0.30
σ_ε	Measurement error SD	1.00

Transition Hazards

Gompertz baseline $\lambda_0(t) = a e^{bt}$, subject-specific LP:

Pre-dialysis (0→1 and 0→2)

$$LP_{h,i}(t) = \alpha_h Z_i + \phi_{h,1} m_i(t) + \phi_{h,2} \dot{m}_i, \quad h \in \{01, 02\}$$

Post-dialysis (1→2, clock reset at initiation time v_i)

$$LP_{12,i}(s) = \alpha_{12} Z_i + \theta_m m_i(v_i) + \psi_d v_i$$

0→1 (Initiation)			0→2 (Pre-dial death)			1→2 (Post-dial death)		
α_{01}	Confounder	0.30	α_{02}	Confounder	0.40	α_{12}	Confounder	0.30
$\phi_{01,1}$	Level	-0.10	$\phi_{02,1}$	Level	-0.05	θ_1	Level at v_i	-0.02
$\phi_{01,2}$	Slope	-2.00	$\phi_{02,2}$	Slope	-1.00	ψ_d	Delay effect	0.05

Study Design and Event Rates

Design parameters:

Setting	Value
visits	12
Horizon T	34 months
Censoring	$10 + \text{Exp}(0.05)$
Measurement interval	$3 \pm \text{Unif}(-1, 1)$ months

Pilot cohort ($n = 400$):

Event	Rate
Dialysis and death	20.75%
Censored after dialysis	13.75%
Death before dialysis	20.75%
Censored before dialysis	44.75%

Joint Model Specification

Complete-data log-likelihood

$$\ell_c(\theta) = \sum_{i=1}^N \left[\log p(Y_i | \mathbf{b}_i, \theta_{\text{LMM}}) + \log p(T_i, \delta_i | \mathbf{b}_i, \theta_{\text{surv}}) \right]$$

Baseline hazard: cubic B-spline baseline hazard with 6 interior knots.

16 regression parameters estimated via numerical differentiation:

$$\underbrace{\beta_0, \beta_1, \gamma, \sigma_e^2, \Sigma_D}_{\text{LMM}}, \underbrace{\alpha_{01}, \phi_{01,1}, \phi_{01,2}}_{0 \rightarrow 1}, \underbrace{\alpha_{02}, \phi_{02,1}, \phi_{02,2}}_{0 \rightarrow 2}, \underbrace{\alpha_{12}, \theta_m, \psi_d}_{1 \rightarrow 2}$$

Standard errors: Estimated by bootstrap method ($B = 50$ replicates).

Run configuration:

Parameter	Value
Sample size N	400
Replicates R	1000
Bootstrap B	50

Fisher-Scoring tuning:

Parameter	Value
GH nodes	20^2
GL nodes	20
Max Fisher-scoring iter	50
Numerical Differentiation step	10^{-6}

Simulation Results

Model	Param	True	Bias	ESD	ASD	CP
LMM	β_0	40.00	-0.006	0.087	0.099	0.966
	β_1	-1.00	0.014	0.035	0.035	0.931
	γ	-2.00	0.004	0.129	0.150	0.972
0→1	α_{01}	0.30	0.031	0.292	0.340	0.972
	$\phi_{01,1}$	-0.10	0.002	0.060	0.071	0.967
	$\phi_{01,2}$	-2.00	-0.231	1.266	1.465	0.974
0→2	α_{02}	0.40	-0.003	0.268	0.343	0.982
	$\phi_{02,1}$	-0.05	-0.005	0.060	0.080	0.981
	$\phi_{02,2}$	-1.00	0.026	1.020	1.391	0.981
1→2	α_{12}	0.30	0.032	0.300	0.454	0.992
	θ_m	-0.02	-0.003	0.035	0.053	0.985
	ϕ_d	0.05	-0.001	0.041	0.069	0.992

- All replicates are converged and all coefficients have low bias (relative $\leq 15\%$).
- Bootstrap standard error for survival parameters are higher than empirical standard error, leading to significantly higher CP.
- Need to figure out the reason, maybe the bootstrap sample is not enough.

Summary

Strengths of JM ECM:

- LMM parameters: negligible bias, matches LME
- Alpha coefficients (α_{01} , α_{02} , α_{12}): well-identified, CP ≥ 0.90
- $\phi_{02,2}$ bias reduced 32% vs two-stage
- ΔRRMST bias reduced 24%; ESD reduced 14%
- Correct theoretical framework for competing risks

Remaining issues:

- All 20 ECM runs **did not converge** (50-iter cap)
- Profile SEs 4–6 $\times$ too small $\Rightarrow$ CP collapses for phi/theta
- $\phi_{02,2}$ bias still -0.41 ; structural sparse-data limit at $N = 300$
- θ_2 identified but with large ESD (0.82)
- CDR slightly lower than two-stage (0.574 vs 0.589)

Key take-away

The ECM joint model improves point estimation and ΔRRMST accuracy relative to the frozen-landmark two-stage estimator, but reliable inference (coverage) requires full convergence with production ECM settings

Next Steps

- ❶ **Re-run with production settings** (`jm_max_iter=100`, `jm_bfgs_maxit=20`, `K_quad=9`, `R=30`):
 - Verify ECM convergence ($\max |\Delta \theta| < 10^{-5}$)
 - Check that ASD $\approx$ ESD for phi/theta parameters
 - Obtain valid CP
- ❷ **Understand $\phi_{02,2}$ bias in the JM:**
 - Run a JM oracle check with true $\mathbf{b}_i$ used for initialization (not two-stage Cox)
 - Verify whether bias decreases at $N = 1000$
- ❸ **Five-scenario expansion** (Scenarios 1–5):
 - Sc1 (flat trajectory): JM vs two-stage should agree
 - Sc2 (dynamic, base): main comparison scenario
 - Sc3 (high measurement error): test JM robustness to BLUP noise
 - Sc5 (unmeasured frailty): JM partially absorbs via $\mathbf{b}_i$?
- ❹ **Bootstrap CI for JM** ($B = 200$): compare bootstrap SE to profile SE

- In the initial simulation, $H_i(t_k)$ is represented by $Z_i, m_i(t_k)$, and $m'_i(t_k)$. In real data, $H_i(t_k)$ should also include symptoms, hospitalization, potassium, bicarbonate, volume overload, comorbidity, and access status when available.
- The 'no-peeking' guarantee in slide 21 applies to the updating of individual random effects (b_i) at time u . Population parameters (θ) are estimated using the full cohort data to maximize structural stability; this includes longitudinal measurements that may extend past u for subjects who have not yet transitioned. In a strict 'landmark-respecting' paradigm, these could be estimated using only data prior to u to avoid all future leakage, though this generally sacrifices parameter efficiency.
- Care must be taken in the simulation design: with a low random-slope variance (e.g., $SD=0.12$), the joint model may face challenges in identifying the association parameter ϕ separately from the random-slope variance. For base scenario calibrations, increasing the random-slope variance is recommended to ensure sufficient signal for the model to distinguish the latent trajectory dynamics from the hazard association.